

- 左边一个红蓝表是从100个回答里面总结的
- 右边一个红蓝表是NASA的危险定级表
- 左边合理

### Background | Maintenance Data Mining

Maintenance data, documented by technicians during O&M activities, provides a comprehensive understanding of system lifelong failures and maintenance.

- System configuration;
- Failure mechanisms;
- Failure criticality;
- Maintenance actions;
- O&M insights;

This information is instructive for subsequent O&M actions, leading to various knowledge mining and management tools:

- Reliability analysis;
- FTA, FMEA, FMECA;

Traditional methods analyze maintenance data in manual...  
The big data environment challenges the conventional...  
checking-based methods.

Some companies waive to analyze these data due to the...  
possibly labor cost resulting from...

| Maintenance Data Mining     |                          |                          |                          | NASA Hazard Rating Table |            |               |         |
|-----------------------------|--------------------------|--------------------------|--------------------------|--------------------------|------------|---------------|---------|
| Failure Mode                | Failure Mechanism        | Failure Consequence      | Failure Criticality      | Hazard Rating            |            |               |         |
|                             |                          |                          |                          | Severity                 | Occurrence | Detectability | Overall |
| 1. Engine failure           | Engine failure           | Engine failure           | Engine failure           | 5                        | 12         | 5             | 60      |
| 2. Fuel system failure      | Fuel system failure      | Fuel system failure      | Fuel system failure      | 4                        | 8          | 12            | 48      |
| 3. Landing gear failure     | Landing gear failure     | Landing gear failure     | Landing gear failure     | 3                        | 9          | 3             | 27      |
| 4. Hydraulic system failure | Hydraulic system failure | Hydraulic system failure | Hydraulic system failure | 2                        | 6          | 6             | 36      |
| 5. Brake system failure     | Brake system failure     | Brake system failure     | Brake system failure     | 1                        | 3          | 3             | 9       |